SENSOR SUBSYSTEM PCB — ELECTRICAL POWER BUDGET & DISSIPATION ANALYSIS

System Rail: +5.0V DC (U1 L7805) | Primary Input: 9.0V – 12.0V DC (Vin) | Primary Protection: 500mA Fuse (F1) | Status: Validated

MAX +5V RAIL CURRENT	MAX +5V RAIL POWER	MAX REGULATOR DISSIPATION	PEAK VIN INPUT CURRENT
34.86 mA	**174.30 mW**	**315.98 mW**	**40.86 mA**

TABLE 1: SUBSYSTEM POWER BUDGET (+5V REGULATED RAIL)

Ref Des	Component / Part Number	Function / Circuit Node	Supply Voltage (V)	Typical Current (mA)	Max Current (mA)	Typical Power (mW)	Max Power (mW)	Notes / Operating Condition
U2	Microchip PIC18F57Q43 (DMG_PIC18F57Q1)	8-Bit MCU (Core + ADC, I2C, UART)	5.0	8.00	16.00	40.00	80.00	Internal HFINTOSC active; GPIO LED load counted separately in D1
U3	TI LM324N (DMG_LM324N1)	Quad Op-Amp Signal Conditioning	5.0	1.00	2.00	5.00	10.00	All 4 channels biased; driving high-impedance PIC ADC inputs
U4	Honeywell HIH6130-021-001 (DMG_HIH6130-021-1)	Digital Humidity / Temp Sensor (I2C)	5.0	0.65	1.00	3.25	5.00	650 uA active measurement conversion; 1 uA in sleep mode
U5, U6	Analog Devices TMP36GT9Z (x2) + R1 (10k)	Analog Temp Sensors (DMG_TMP36GT9Z1/2)	5.0	0.20	0.30	1.00	1.50	50 uA quiescent per sensor (x2) + 0.10 mA load through R1 (10k)
R_LDR1, R2	Luna PDV-P8103 (DMG_PDV-P1) + R2 (10k)	Photoresistor Voltage Divider (LIGHT_IN)	5.0	0.20	0.50	1.00	2.50	Typical indoor light (~15k LDR + 10k R2); Max in bright light (5 V / 10 kOhm)
R5, R6	2.2k 1/4W Pull-Up Resistors (x2)	I2C Bus Pull-Ups (I2C_SCL, I2C_SDA)	5.0	1.14	4.55	5.70	22.75	2.27 mA per line when pulled low (5 V / 2.2 kOhm); ~25% active duty cycle
SW1, R7	Tactile Push Button + 10k 1/4W Pull-Up	User Input & Hardware Debounce (BTN1_RC4)	5.0	0.05	0.50	0.25	2.50	0 mA unpressed; 0.50 mA peak when button is held down (5 V / 10 kOhm)
D1, R8	Green Debug LED + 330R 1/4W Resistor	Status Indicator (DBG_LED on RC2)	5.0	9.09	10.00	45.45	50.00	Assumes VF = 2.0 V LED drop: (5.0 V - 2.0 V) / 330 Ohm = 9.09 mA
C1-C7	Ceramic Decoupling / Filter Capacitors	Rail Bypass & Signal Filtering	5.0	0.00	0.01	0.00	0.05	Negligible DC leakage current at steady state
TOTAL (+5V)	**All +5V Rail Loads Combined**	**Regulated +5V Bus (U1 Output)**	**5.0**	**20.33 mA**	**34.86 mA**	**101.65 mW**	**174.30 mW**	**Total load drawn from L7805 output pin 3**

TABLE 2: PRIMARY INPUT (VIN), REGULATOR DISSIPATION & FUSE MARGIN

Parameter	Formula / Source	Typical Value (at 9V Vin)	Max Value (at 9V Vin)	Max Value (at 12V Vin)	Limit / Rating	Safety Margin
+5V Load Current (I_load)	*Sum of +5V branch currents*	20.33 mA	34.86 mA	34.86 mA	1000 mA (L7805 max)	96.5% margin
U1 (L7805) Quiescent Current (I_Q)	*L7805 datasheet ground pin current*	4.20 mA	6.00 mA	6.00 mA	8.0 mA max spec	Normal operating range
Total Input Current (I_in)	*I_in = I_load + I_Q*	24.53 mA	40.86 mA	40.86 mA	500 mA (F1 Fuse)	12.2x below fuse rating
Regulator Power Dissipation (P_reg)	*(V_in - 5 V) * I_load + V_in * I_Q*	119.12 mW	193.44 mW	315.98 mW	~1500 mW (TO-220 no heatsink)	Cool operation (no heatsink required)
Total Board Input Power (P_in, total)	*V_in * I_in*	220.77 mW	367.74 mW	490.32 mW	4500 mW (500mA at 9V)	91.8% power margin